

Manipulation of Formula Worked Examples - Basic

Example 1

Make d the subject of the formula:

$$b = c - d$$

$$c - d = b \quad \text{Rearranging formula}$$

$$-d = b - c \quad \text{Subtract } c \text{ from both sides of the formula}$$

$$d = -b + c \quad \text{Multiply across by } -1$$

$$d = c - b \quad \text{Rearrange so positive elements come first}$$

Example 2

Consider that $F \neq 0$. Solve for C :

$$F = \frac{9}{5}c + 32$$

$$\frac{9}{5}C + 32 = F$$

$$\frac{9}{5}C = F - 32 \quad \text{Subtract 32 from both sides of the formula}$$

$$9C = 5(F - 32) \quad \text{Multiply both sides by 5}$$

$$C = \frac{5}{9}(F - 32) \quad \text{Divide both sides by 9}$$